Acoustic Variations in Child-Directed Speech Facilitate Referent Label Comprehension

Bhuvana Narasimhan[1], Rebecca Scarborough, Allison Hilger, Kanupriya Kale, Justin Bai, Chloe Circenis, Tessa Moskoff, Alex Hutton, Karis Lowe

University of Colorado Boulder

Background. Across different cultures, caregivers adapt the acoustic features of their speech (e.g., pitch changes, vowel lengthening) when addressing children, which can facilitate emotion regulation, guide attention, and support language learning [1-6]. But the features of child-directed speech (CDS) may also have a communicative motivation, to help listeners process information that is difficult to comprehend [7]. Prior work shows that English-speaking mothers increase pitch in CDS relative to adult-directed speech (ADS) and lengthen vowels when referent identification is more difficult (e.g., in corrective instructions) particularly in CDS [8]. But it is not known if–and under what conditions–these kinds of maternal speech modifications actually facilitate referential success. We hypothesized that listeners will better comprehend labels produced in CDS vs. ADS, especially in contexts where referent identification is relatively difficult. We also expected these effects to be greater in children, who might benefit more from acoustic features in CDS that enhance clarity.

In **our study**, we employed a referent identification task building on prior research with adults [9]. Child (N=31) and adult (N=23) participants who were speakers of English were shown arrays of four pictures on a tablet screen and asked to touch the target referent on hearing the instruction in "Touch the X". The pictures consisted of the target referent ('dog'), a semantically related competitor ('bone'), a phonetically similar competitor ('doll'), and an unrelated competitor ('cake') (Figure 1). The instructions containing the target referent label were produced by mothers addressing a hypothetical child (CDS) or adult listener (ADS). Each target label was produced in 4 contexts that increased in difficulty of referent identification in the following order: the target referent was presented in isolation ('single' context), presented together with a competitor with a phonetically unrelated label ('non-confusable' context), presented together with a competitor with a phonetically related label ('confusable' context), or involved correction of a prior referential choice ('corrective' context) (Figure 2). There were 18 target labels produced by a range of maternal talkers in each of the 2 speech type conditions and each of the 4 referent conditions.

Results: Mixed effects regression analyses examined the effects of participants' age (child, adult), speech type (CDS, ADS), and referential context (single, non-confusable competitor, confusable competitor, corrective) on reaction times for correct target referent choice (measured from target label offset) and accuracy. The expected three-way interaction of speech type, referential context, and age was not significant, but speech type interacted with referential context (β = -0.135, t-value = -2.465, $p < .05$): listeners responded faster to targets from CDS versus ADS when they were produced in corrective contexts versus less difficult (single, non-confusable competitor, confusable competitor) contexts (Figure 3). Speech type also interacted with age (β = 0.083, t-value = 2.169, $p < .05$): adult listeners responded faster to targets from CDS versus ADS compared to children (Figure 4). Adults were also more accurate than children overall (β = 2.200, t-value = 7.069, $p < .001$); neither speech type nor referential context influenced accuracy.

Discussion: Our findings show that listener comprehension is affected by acoustic modifications in maternal speech based on the type of addressee to whom the speech is directed as well as the referential context in which it is produced. Target referent labels that are produced in contexts where mothers modify their speech the most–when producing corrective feedback directed to children–are also understood most easily. Counter to our expectation, adults, rather than children, process information more quickly in CDS than ADS. This finding is surprising but may be explained if adults are more practiced than children at exploiting acoustic features in speech to identify referents. Current theorizing is divided regarding the benefits of CDS in different domains of child language development [10]. By comparing the effects of CDS-ADS differences in varying referential contexts on listener comprehension, the present study extends understanding of how acoustic modifications in speech can facilitate referential communication in children and adults.

[1] Corresponding author's email: narasimb@colorado.edu

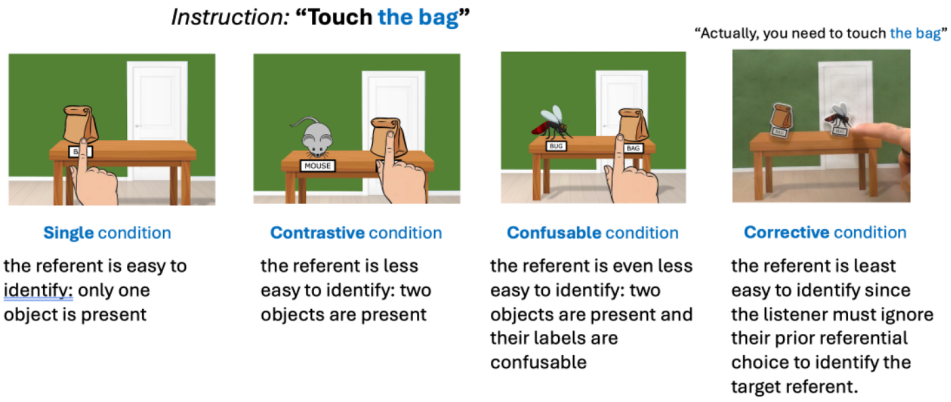

Figure 1. Sample Trial in Referent. Task

Figure 2. Target labels produced by mothers in scenes varying in Identification ease of referent identification

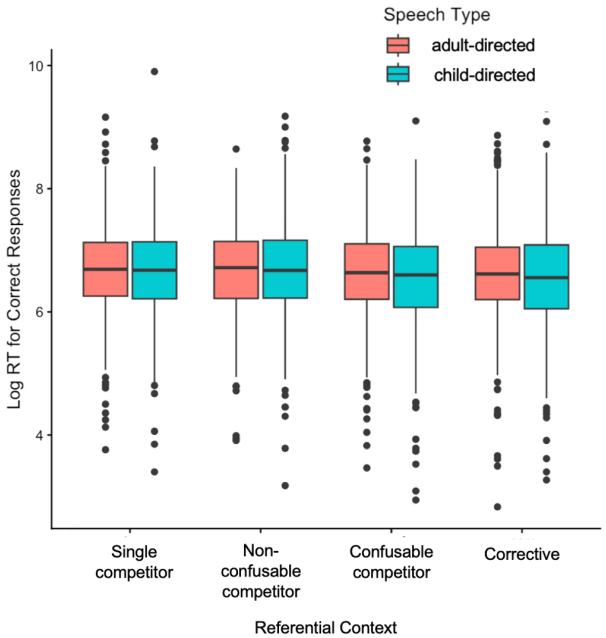

Figure 3. Mean RT by Referential Context and Speech Type

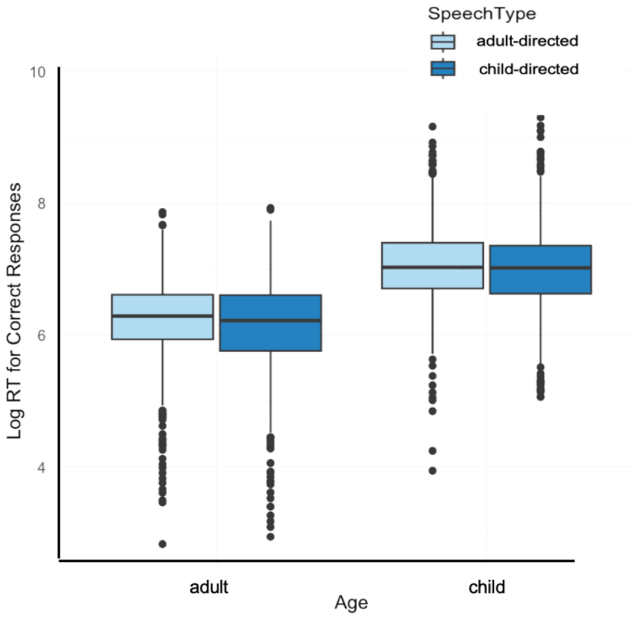

Figure 4. Mean RT by Age and Speech type

References

[1] Gleason, J.B. & N. Bernstein Ratner (eds.). 2023. The Development of Language (10th ed.). Plural Publishing. [2] Cychosz M, Edwards JR, Bernstein Ratner N, Torrington Eaton C, Newman RS. 2021. Acoustic-Lexical Characteristics of Child-Directed Speech Between 7 and 24 Months and Their Impact on Toddlers' Phonological Processing. Frontiers in Psychology. 12:712647. [3] Schick J, Fryns C, Wegdell F, Laporte M, Zuberbuhler K, van Schaik CP, et al. 2022 The function and evolution of child-directed communication. PLoS Biol 20(5): e3001630. [4] Fernald, A. 1989. Intonation and communicative intent in mothers' speech to infants: Is the melody the message? Child Development, 60(6), 1497-1510. [5] Fernald, A. 1991. Prosody in speech to children: Prelinguistic and linguistic functions. In R. Vasta & R. Vasta (Eds.). Annals of child development (Vol. 8, pp. 43–80). London, England: Jessica Kingsley Publishers. [6] Saint-Georges, C., Chetouani, M., Cassel, R., Apicella, F., Mahdhaoui, A., Muratori, F., Cohen, D. 2013. Motherese in interaction: At the cross-road of emotion and cognition? (A systematic review). PLoS ONE, 8(10), e78103. [7] Fernald A, Mazzie C. 1991. Prosody and Focus in Speech to Infants and Adults Developmental Psychology. 27: 209-221. [8] Author 1 et al. [9] Author 2 et al. [10] Kempe, V., Ota, M., & Schaeffler, S. (2024b). Does child- directed speech facilitate language development in all domains? A study space analysis of the existing evidence. Developmental Review, 72, 101121.